

AMAL JOY

ROBOTICS AND AUTOMATION ENGINEER

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CAREER SUMMARY

Accomplished Automation Engineer with expertise in automation and robotics, experienced in delivering control system solutions for industrial and aerospace applications. Demonstrates strong capability in executing tasks while adhering to stringent safety standards for both personnel and equipment. Worked on automation projects at ISRO – SDSC SHAR, supporting critical process systems for launch vehicle programs. Adept at collaborating with engineers and technicians, with a strong understanding of process requirements, system testing, and troubleshooting activities.

CORE COMPETENCIES

- PLC – Proficient in Siemens S7-1200, S7-1500, S7 400, S7 300, S7 200, Allen Bradley CompactLogixL33R, Micrologix 1400 and 1500, Schneider M221 PLCs and experienced on Schneider Modicon M340 and M580 PACs.
- Proficient in Programing languages like STL, SCL, LAD, FBD, SFC for process automation and Python, VBScript, C programing, and Java for data processing, SCADA customization, and engineering application.
- SCADA, HMI & Historian Systems – Skilled in AVEVA InTouch, Plant Scada, AVEVA Edge, Vijeo Designer, Simatic WinCC, WinCC Unified, WinCC OA, Factory Talk View and Ignition Scada for industrial monitoring, control, alarm management and visualization applications.
- System Integration and Troubleshooting – Experience in Industrial Automation system, SCADA lifecycle support, commissioning, troubleshooting, and operator support activities.
- DATABASE & OS Knowledge – Familiar with Microsoft Windows environments, SQL Server, Data Logging and Alarm, Trend management System.
- TIA Portal – Advanced programming and configuration using Siemens TIA Portal for PLC and HMI systems
- Industrial Protocols – Knowledgeable in Modbus RTU, TCP/IP, Ethernet/IP, OPC UA, HART, PROFIBUS, and PROFINET, MQTT, IEC 61850, SNMP.
- VFD & Servo Configuration – Setup and tuning of Variable Frequency Drives Schneider Altivar 312,31 and Siemens G20 and Delta ASDA Soft servo drives.
- Distributed Control System (DCS) – Practical experience with Honeywell C300 DCS for process automation, centralized monitoring, and control in industrial setups.

PROFESSIONAL EXPERIENCE

AUTOMATION ENGINEER | Master Systems LLC Sept 2026 – present

- Working on Marine Automation projects, Delivering reliable IAS, PMS and Navigation Safety systems for commercial vessels and offshore assets.

PROJECT ENGINEER | Proficient Automation and Controls Ltd. Nov 2025 – Aug 2026

- Working on automation projects at SDSC SHAR – ISRO, supporting mission-critical industrial and aerospace control systems.
- Developed PLC program logic for automatic sequential control of solid propellant fuel casting systems and Bowlcleaning after casting that is used in GSLV and Gaganyaan programs.

AUTOMATION ENGINEER | SMEC Automation Pvt Ltd. Oct 2024 – Oct 2025

- Designing and programming PLC systems for Marine and Industrial automation projects.
- Developing SCADA systems and integrating real-time communication using protocols like Modbus RTU, TCP/IP, Ethernet/IP, OPC UA, Profibus.
- Interfacing and calibrating field instruments include pressure transmitters, control valves, and PID controllers.
- Support system installation, wiring, troubleshooting.

PROJECTS

BOWL CLEANING FACILITY ISRO - SDSC SHAR

- Designed PLC logic for automatic bowl cleaning sequence used in propellant casting preparation processes.
- Used Siemens S7-1500 PLC for control logic development, programming in STL, SCL and system integration.
- Integrated SIMOCODE pro motor management system for motor control, protection, and diagnostics within the automation system.
- Implemented hydraulic pump control using dedicated drive/control modules -Parker card, including Start/Stop, Flow control, Pressure Control and PID Tuning.
- Integrated a field-mounted HMI for local operation, status indication, and manual control of motors and hydraulic systems.
- Designed WinCC SCADA screens for process visualization, status monitoring, alarms, and operator control.

CASTING FACILITY ISRO - SDSC SHAR

- Developed PLC program logic for fully automatic sequencing of solid propellant fuel casting processes for GSLV and Gaganyaan launch vehicle programs.
- Implemented process control logic in STL programming for safe boiler startup, shutdown, and interlocks.
- Designed Redundant SCADA system for real-time monitoring and operator interface, ensuring reliable control of pressure, temperature, and alarms.
- Worked with Siemens S7-405H redundant PLC system, ensuring high availability and fail-safe operation.

ALARM MONITORING SYSTEM – HYBRID PROPULSION SYSTEM

- Developed and configured alarms for a Hybrid Propulsion Control System integrating with Siemens PCS7 with S7-405H PLC with Redundancy.
- Implemented alarm monitoring, event handling, alarm acknowledgement, and historical alarm diagnostic for propulsion and auxiliary systems.

BOILER CONTROL SYSTEM

- Developed and programmed a complete boiler control system using Schneider Modicon M340 PLC.
- Implemented process control logic for safe boiler startup, shutdown, and interlocks.
- Designed SCADA system for real-time monitoring and operator interface, ensuring reliable control of pressure, temperature, and alarms.

INERT GAS GENERATOR

- Designed SCADA system in Siemens WinCC for monitoring and control of inert gas generator processes.
- Integrated process parameters including gas flow, pressure, and safety interlocks into graphical interface.
- Enhanced operator visibility and safety compliance through alarm handling, trend logging, and reporting features.

CALORIFIER CONTROL SYSTEM

- Designed an automated hot water control system using Schneider M221 PLC.
- Programmed temperature-based logic for heating control and process reliability.
- Designed AVEVA InTouch SCADA applications for process visualization, Alarm Management, Operator Control and Real time system monitoring.

EDUCATION

B. TECH | ROBOTICS AND AUTOMATION | Kerala Technical University

2020 – 2024

CERTIFICATIONS

- Post Graduate Diploma in Industrial Automation
- Advanced Diploma in Distributed Control System – Honeywell C300
- Ignition 8.1 Credential – Inductive Automation
- Google IT Automation with Python - Coursera